

ASG Airlines – End-to-End Data Engineering

Aim

The objective of this study was to develop an end-to-end data engineering pipeline for ASG Airlines to clean, validate, transform, store, and analyze flight, booking, passenger, and payment data.

Methodology:

The raw Excel data was first analyzed and inspected for any problems such as missing values, duplicates, inconsistencies, invalid entries, or data relationship problems. The data was then cleaned and transformed with Python and Pandas. Flight duration and timestamp were validated, including an issue of timestamp for an overnight flight. Missing airline data was inferred from flight ID prefix validation. Passenger age was computed from date of birth. Any data quality problem was noted instead of ignoring it silently.

The cleansed data was put into MySQL database, where SQL views were created for airline distribution, booking status, flights, route traffic, payment statistics, and general KPIs. Finally, the Power BI dashboard was developed from these views.

Tech Stack

- Python & Pandas – data ingestion, cleaning, transformation and validation
- MySQL – structured data storage and SQL-based analysis
- SQL – analytical views and KPI generation
- Power BI – dashboard and visual reporting
- Jupyter Notebook – data exploration and quality analysis
- Excel – source data

Azure was recommended in the case study, but due to the lack of Azure availability in the working environment, the pipeline was implemented locally using Python, MySQL, SQL and Power BI.

Key Findings

The cleansed data set included 1,005 flight entries, 1,000 bookings, 1,039 passengers' details and 1,000 payment details. The following data problems were addressed: duplicates, missing and wrong data, discrepancies in airlines, passengers' ages or dates of birth, incorrect payments and the issue with the time

stamp for flights. Sensitive passenger details were also anonymized within the analytics-ready data set of passengers.

Flight duration average was 164.62 minutes and was shown via the Power BI dashboard along with airline distribution, booking statuses, traffic on routes and means of payment.